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Assignment 1: Token Recognition using C/C++

AIM

To write a C/C++ program that recognizes and classifies tokens such as keywords, identifiers, operators, numbers, and special characters from a given source program.

ALGORITHM

Lexical analysis is the first phase of a compiler, where the source code is scanned and converted into meaningful units called tokens.

Algorithm

1. Define a list of C keywords.
2. Read a line or multiple lines of source code as input.
3. Traverse the input character by character.
4. Group characters to form tokens:
 - Keyword: If token matches predefined keyword list
 - Identifier: Alphabetic strings not keywords
 - Number: Sequence of digits
 - Operator: Symbols like + - * / =
 - Special Character: Symbols like ; , () { }
5. Print each token along with its corresponding type.

C++ PROGRAM

```
#include <iostream>
#include <cstring>
#include <cctype>
using namespace std;

bool isKeyword(string word) {
    string keywords[] = {
        "int", "float", "double", "char", "if", "else", "while",
        "for", "return", "void", "break", "continue"
    };

    for (string k : keywords) {
        if (word == k)
            return true;
    }
    return false;
}

bool isOperator(char ch) {
    string operators = "+-*/=%";
    return operators.find(ch) != string::npos;
}

bool isSpecialChar(char ch) {
    string special = ";,(){}[]";
    return special.find(ch) != string::npos;
}

int main() {
    string input;
    cout << "Enter source code line:\n";
    getline(cin, input);

    for (int i = 0; i < input.length(); i++) {

        // Skip spaces
        if (isspace(input[i]))
            continue;
    }
}
```

```

// Identifier or Keyword
if (isalpha(input[i])) {
    string word = "";
    while (i < input.length() && (isalnum(input[i]) || input[i] == '_')) {
        word += input[i];
        i++;
    }
    i--;

    if (isKeyword(word))
        cout << word << " : KEYWORD\n";
    else
        cout << word << " : IDENTIFIER\n";
}

// Number
else if (isdigit(input[i])) {
    string number = "";
    while (i < input.length() && isdigit(input[i])) {
        number += input[i];
        i++;
    }
    i--;
    cout << number << " : NUMBER\n";
}

// Operator
else if (isOperator(input[i])) {
    cout << input[i] << " : OPERATOR\n";
}

// Special Character
else if (isSpecialChar(input[i])) {
    cout << input[i] << " : SPECIAL CHARACTER\n";
}
}

return 0;
}

```

OUTPUT

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  PLAYWRIGHT

vanshikadeswal@Vanshikas-Laptop compiler lab % clang++ lab1.cpp -o lab1
./lab1

clang: error: no input files
zsh: no such file or directory: ./lab1
● vanshikadeswal@Vanshikas-Laptop compiler lab % clang++ lab1.cpp -o lab1_compiler
./lab1_compiler

clang: error: no such file or directory: 'lab1.cpp'
clang: error: no input files
Enter source code line:
int a = b + 10;
int : KEYWORD
a : IDENTIFIER
= : OPERATOR
b : IDENTIFIER
+ : OPERATOR
10 : NUMBER
; : SPECIAL CHARACTER
○ vanshikadeswal@Vanshikas-Laptop compiler lab %
```

RESULT

The program successfully identifies and classifies keywords, identifiers, operators, numbers, and special characters from the given source code, demonstrating basic lexical analysis functionality.